blades are available in various sizes from 100 mm to 400 mm. The revolution of the blade will be around 1000 RPM and are covered with plastic or metallic grill.

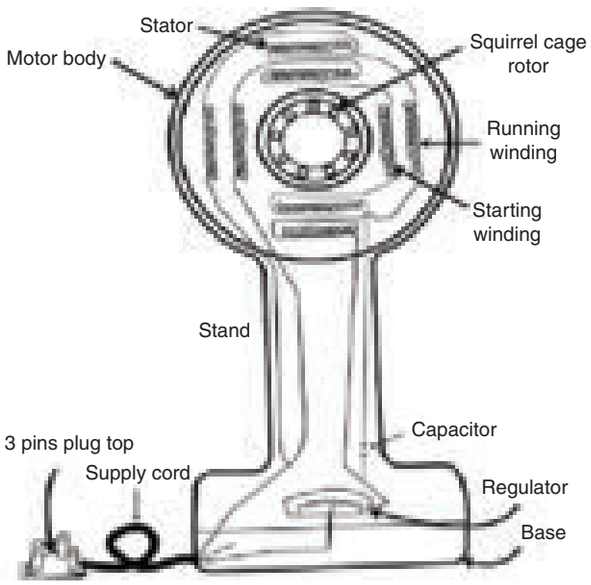



Fig. 4.16 **Table fan**

## 4.2.3 Exhaust fan



Fig. 4.17 **Exhaust fan**

The exhaust fan is as shown in figure 4.17, is used to exhaust the unwanted air present inside rooms, cinema theatres, marriage halls, factories, homes, industries, kitchens and toilets.

### a) Construction

#### i. Stator

The stator is made up of a silicon steel plate and is small. Both the primary and secondary coils are located with $90^0$ electric angles.

#### ii. Rotor

Squirrel cage type rotor is used in this exhaust type of fan.

#### iii. Blades

The blades are made up of a steel or aluminum sheet. It contains three or four blades. The blades are fixed on the shaft and fitted with screws.

### b) Working principle

Its structure is similar to that of table fan. Capacitor is not used in this type. This fan exhausts heats produced during the summer season, creating low pressure inside of the room and causes cool air to enter in. The sweep of the fans are available from 230 mm to 380 mm.

### Trouble shoot chart of an electric fan

| Sl. No. | Reasons | Defects | Remedies |
|---|---|---|---|
| 1. | The fan is not rotating | The connection may be loose or in open circuit. | Connections are tested with test lamp and open circuit in it is rectified. |
| | | Open circuit or short circuit may be in the coils. | Check the coils with test lamp and change the coil, if necessary. |
| | | Fault in capacitor | Change the capacitor. |



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